

CS & IT Engineering



**Lecture Number
03**



By- Ankit Doyla Sir



Topics to be Covered

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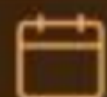
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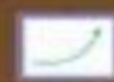


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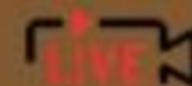


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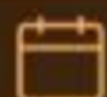
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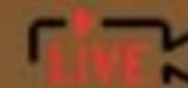


CSIT



Batch A

Live



Substring

(Contiguous sequence of symbols within a string)

If 'u', w be the two string defined over the alphabet (Σ) then 'u' is said to be substring of 'w' If 'u' is obtained from w as it is in the same order.

Note:

- ❑ If u is a substring of w then $|u| \leq |w|$
- ❑ Every string is a substring to itself and empty string ϵ is the substring of every string

1. $W = \text{TOC}$

$$\text{Substring} = \{ \epsilon, T, O, C, TO, OC, TOC \}$$

2. $W = \text{GATE}$

$$\text{Substring} = \{ \epsilon, G, A, T, E, GA, AT, TE, GAT, ATE, GATE \}$$

3. $W = \text{ANKIT}$

$$\text{Substring} = \{ \epsilon, A, N, K, I, T, AN, NK, KI, IT, ANK, NKI, KIT, ANKI, NKIT, ANKIT \}$$

Type of Substring



1. Trivial substring
2. Non trivial Substring



Trivial substring

Let w is any string defined over the alphabet Σ , then the substring w itself and empty string ϵ is called as trivial substring

1. $w = \text{TOC}$

$$\text{Trivial substring} = \{ \text{TOC}, \epsilon \}$$

2. $w = \text{GATE}$

$$\text{Trivial substring} = \{ \text{GATE}, \epsilon \}$$



Non-Trivial substring

Let w is any string defined over the alphabet Σ , then any substring of w other than the substring w itself and empty string ϵ , is called as Non trivial substring

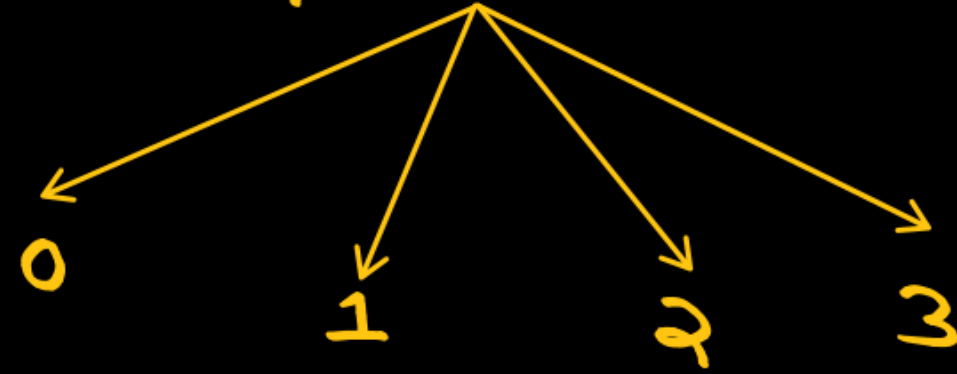
1. $w = TOC$

$$\text{Non trivial substring} = \{ T, O, C, TO, OC, \}$$

2. $w = GATE$

$$\text{Non trivial substring} = \{ G, A, T, E, GA, AT, TE, \\ GAT, ATE \}$$

1. $w = \text{TOC}$, $|w| = 3$



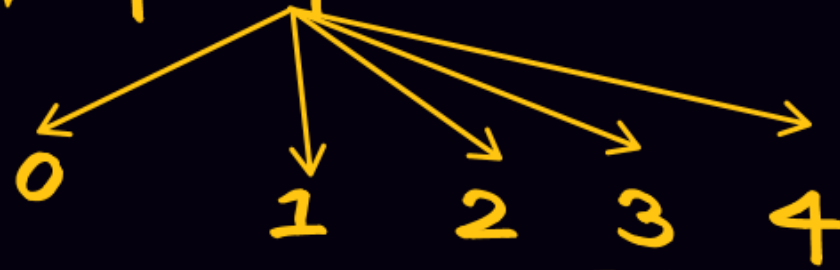
Length

substring

Total

0	→	ε	→	1
1	→	T, O, C	→	3
2	→	TO, OC	→	2
3	→	TOC	→	1
				7 = Σ 3 + 1

2. $w = \text{GATE}$, $|w| = 4$



Length

Substring

Total

0	→	ε	→	1
1	→	G, A, T, E	→	4
2	→	GA, AT, TE	→	3
3	→	GAT, ATE	→	2
4	→	GATE	→	1
				11 = $\sum 4 + 1$

$$|w| = n$$

$$\sum n + 1$$

$$\frac{n(n+1)}{2} + 1$$

If w is any string with distinct symbol and |w|=n

➤ Total number of substring = $\sum_{i=1}^n i + 1 = \frac{n(n+1)}{2} + 1$

➤ Total number of Trivial substring = 2

➤ Total number of Non-Trivial substring = $\sum_{i=1}^n i + 1 - 2 = \sum_{i=1}^n i - 1 = \frac{n(n+1)}{2} - 1$

➤ Total number of Non-Empty string = $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Prefix



A substring with the sequence of starting symbol is called as Prefix.

$$1. w = \text{TOC}$$

$$\text{Prefix} = \{\epsilon, T, TO, TOC\} = 4$$

$$2. w = \text{GATE}$$

$$\text{Prefix} = \{\epsilon, G, GA, GAT, GATE\} = 5$$



Proper Prefix

The prefix without string itself is called as proper Prefix.

1. $w = \text{TOC}$, $|w| = 3$

$$\text{Proper Prefix} = \{ \epsilon, T, TO \} = 3$$

2. $w = \text{GATE}$, $|w| = 4$

$$\text{Proper Prefix} = \{ \epsilon, G, GA, GAT \} = 4$$

Suffix



A substring with the sequence of ending symbol is called as Suffix.

$$1. w = TOC, |w| = 3$$

$$\text{SUFFIX} = \{\epsilon, c, OC, TOC\} = 4$$

$$2. w = GATE, |w| = 4$$

$$\text{SUFFIX} = \{\epsilon, E, TE, ATE, GATE\} = 5$$

Proper Suffix



The Suffix without string itself is called as proper Suffix.

1. $w = \text{TOC}$, $|w| = 3$

$$\text{Proper suffix} = \{ \epsilon, c, oc \}$$

2. $w = \text{GATE}$, $|w| = 4$

$$\text{Proper suffix} = \{ \epsilon, e, te, ate \}$$

If w is any string and $|w|=n$

- Number of Prefix = Number of Suffix = $n+1$
- Number of Proper Prefix = Number of Proper suffix = n
- Trivial substring is common in both Prefix as well as suffix.
- Every prefix or suffix must be a substring but every substring need not be prefix or suffix

EX: $w = TOC$

substring = $\{\epsilon, T, O, C, TO, OC, TOC\}$

PREFIX = $\{\epsilon, T, TO, TOC\}$

SUFFIX = $\{\epsilon, C, OC, TOC\}$

$w = T O C, w = 3$	$w = T O T, w = 3$	$w = T T T$
Prefix = $\{ \epsilon, T, T O, T O C \}$	Prefix = $\{ \epsilon, T, T O, T O T \}$	Prefix = $\{ \epsilon, T, T T, T T T \}$
Suffix = $\{ \epsilon, C, O C, T O C \}$	Suffix = $\{ \epsilon, T, O T, T O T \}$	Suffix = $\{ \epsilon, T, T T, T T T \}$

$$w = T T T T \dots T, |w| = n$$

$$\text{No. of Prefix} = n + 1$$

$$\text{No. of Suffix} = n + 1$$

Substring

1. $w = T O C, |w| = 3$

$$\text{substring} = \{ \epsilon, T, O, C, TO, OC, TOC \} = 7$$

$$\text{Total No. of substring} = \sum_{i=1}^n n+1 = \frac{n(n+1)}{2} + 1$$

2. $w = T O T, |w| = 3$

$$= \frac{3 \times 4}{2} + 1 = 7$$

$$\text{substring} = \{ \epsilon, T, O, TO, OT, TOT \} = 6 \neq \frac{n(n+1)}{2} + 1$$

3. $w = T T T, |w| = 3$

$$\text{substring} = \{ \epsilon, T, TT, TTT \} = 4 \neq \frac{n(n+1)}{2} + 1$$

4. $w = TTTT$, $|w| = 4$

$$\text{substring} = \{ \epsilon, T, TT, TTT, TTTT \} = 5$$

5. $w = TTTTT$, $|w| = 5$

$$\text{substring} = \{ \epsilon, T, TT, TTT, TTTT, TTTTT \} = 6$$

6. $w = TTTT \dots T$, $|w| = n$

$$\text{No. of substring} = n+1$$

$$\neg \text{IF } |w| = n$$

$$n+1 \leq \text{No. of substring} \leq \frac{n(n+1)}{2} + 1$$

Question



Number of trivial substring in MOBILE

A 22

B 21

C 20

D 2

Trivial substring = $\{ \text{MOBILE}, \epsilon \}$

Question



The Number of Substrings and trivial substring in the string "ALGORITHMS" over the alphabet $\Sigma=(A-Z)$

$$n=10$$

A 2, 44

B 44, 2

☒ **C** 56, 2

D 46, 44

$$\begin{aligned}\text{No. of substring} &= \frac{n(n+1)}{2} + 1 \\ &= \frac{5 \times 11}{2} + 1 \\ &= 56\end{aligned}$$

Question



How many substrings (of all lengths inclusive) can be formed from a character string of length n ? Assume all characters to be distinct, prove your answer.

GATE-1989

$$\text{Ans: } \frac{n(n+1)}{2} + 1$$

Question



How many substrings (of all lengths inclusive) can be formed from a character string of length n ? Assume all characters to be same.

Ans: $n+1$

Question



The number of substrings (of all lengths inclusive) that can be formed from a character string of length n is

GATE-1994

A n

C $\frac{n(n-1)}{2}$

B n^2

☒ **D** $\frac{n(n+1)}{2} + 1$

Question

MCR



How many sub strings (non-zero) can be formed from a character string of length n ?

GATE-1998

A n

C 2^n

B n^2

☒ **D** $\frac{n(n+1)}{2}$

(A,D) is correct

Question

GATE-2025



Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, "aa", "bb" or "cc". The number of such strings of length 5 that are possible is _____. (Answer in integer). **Ans: 195**

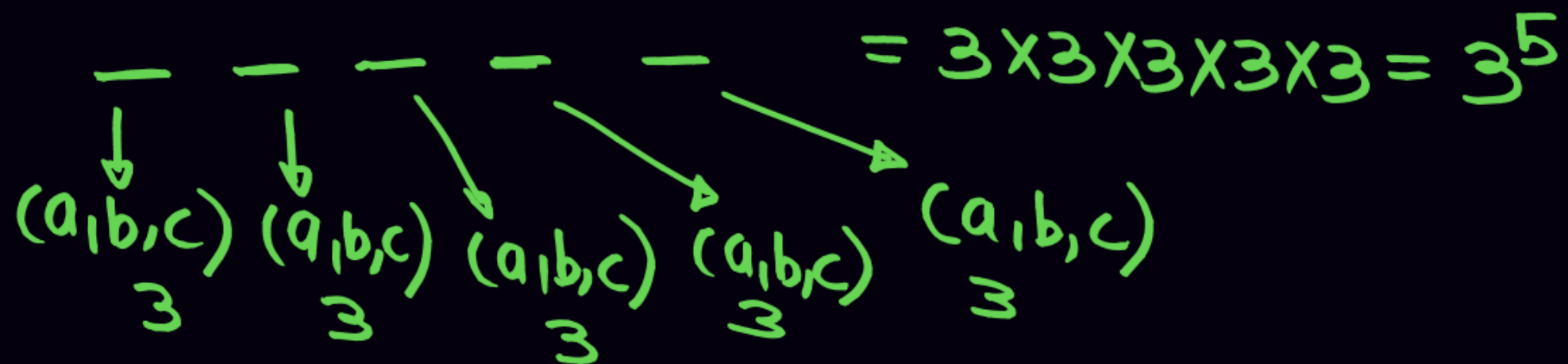
Solⁿ Atleast one occurrence of two consecutive symbol
(length 5)

= Total 5 length string - No occurrence of consecutive symbol

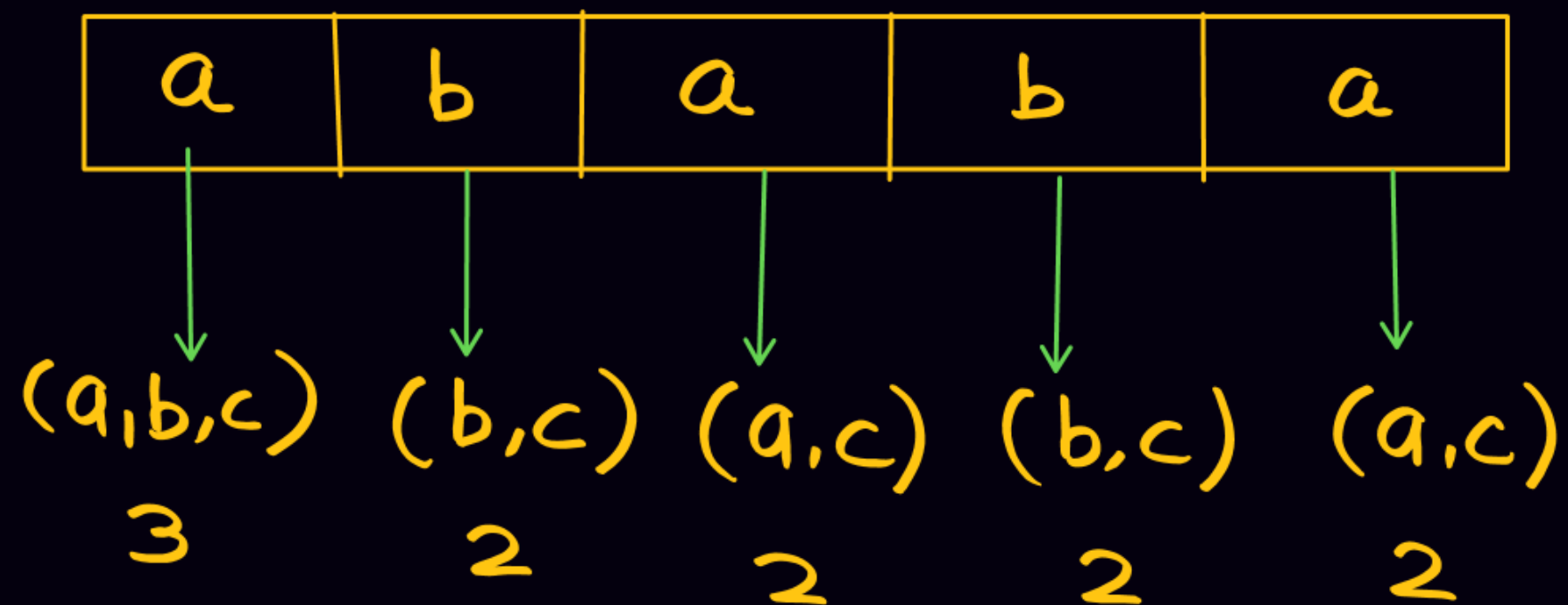
$$= 3^5 - 3 \times 2^4$$

$$= 243 - 48 = 195$$

$$\Sigma = \{a, b, c\}$$



No. occurrence of consecutive symbol



$$= 3 \times 2 \times 2 \times 2 \times 2$$

$$= 3 \times 2^4$$

Q: Atleast one occurrence of two consecutive symbol
(length n) , $\Sigma = \{a, b, c\}$

= Total n length string — No occurrence of consecutive symbol

$$= 3^n - 3 \times 2^{n-1}$$

Question

Let S be the set of all quaternary strings defined over the alphabet $\{a, b, c, d\}$.
Consider all strings in S that contain at least one occurrence of two consecutive
symbols, that is, "aa" , "bb" "cc" or "dd" . The number of such strings of
length 5 that are possible is _____. (Answer in integer).

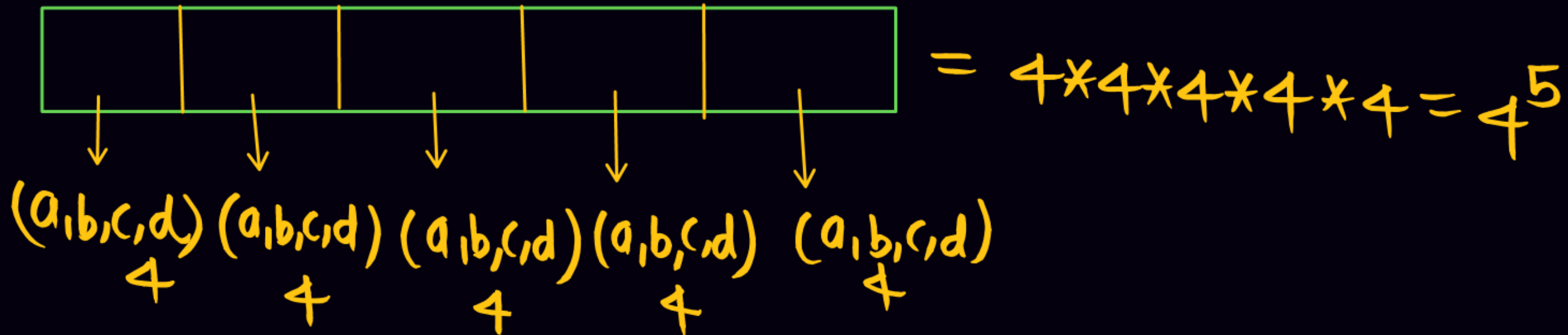
Solⁿ

Atleast one occurrence of two consecutive symbol
(length 5) , $\Sigma = \{a, b, c, d\}$

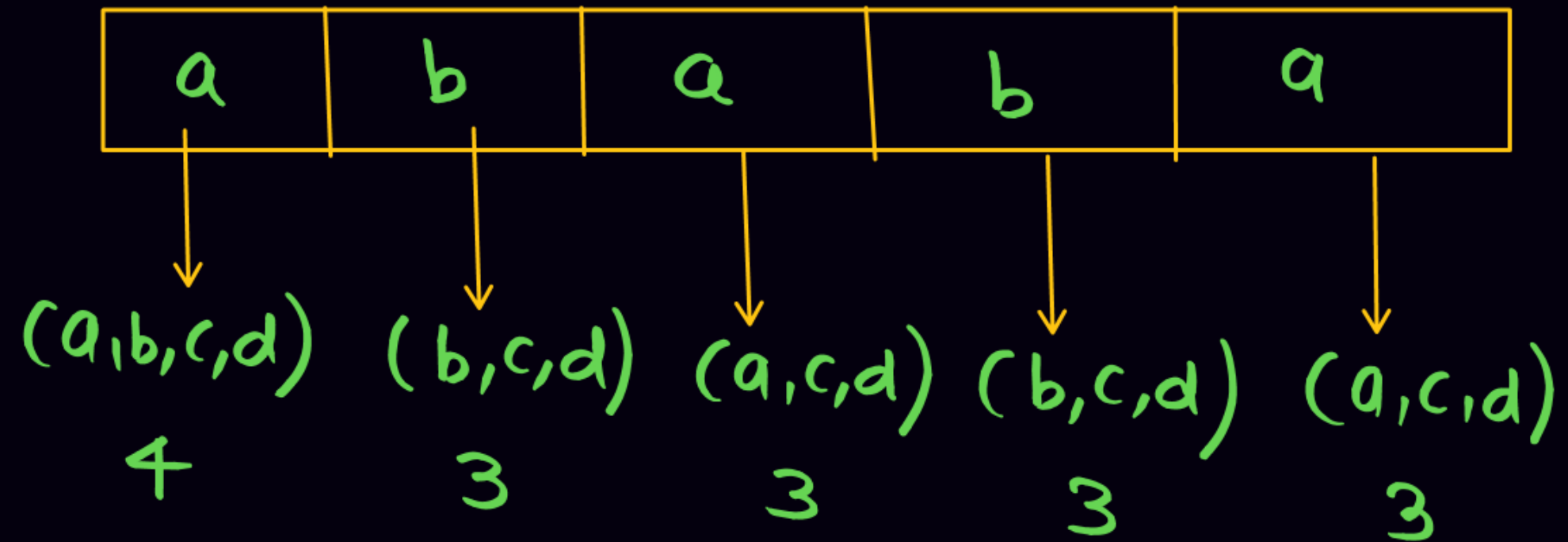
= Total 5 length string - No occurrence of consecutive symbol

$$= 4^5 - 4 \times 3^4$$

$$= 1024 - 324 = 700$$



No occurrence of consecutive symbol $\Sigma = \{a, b, c, d\}$



$$4 \times 3 \times 3 \times 3 \times 3 = 4 \times 3^4$$

Solⁿ

Atleast one occurrence of two consecutive symbol
(length n) , $\Sigma = \{a, b, c, d\}$

= Total n length string - No occurrence of consecutive
symbol

$$= 4^n - 4 \times 3^{n-1}$$

Generalized Formula For Any Alphabet
Atleast one occurrence of two consecutive symbol
(length n)

= Total n length string — No occurrence of consecutive symbol

$$= |\Sigma|^n - |\Sigma| * |\Sigma - 1|^{n-1}$$



Thank You